

Harsh Kumar

POSTDOCTORAL FELLOW IN ASTROPHYSICS

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Personal Details

Date of Birth 13/02/1996 | *Nationality:* Indian

Current role Postdoctoral Fellow, Center for Astrophysics | Harvard & Smithsonian; Junior Investigator, NSF AI Institute for Artificial Intelligence & Fundamental Interactions (IAIFI).

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Research Interests

Time-domain and multi-messenger astrophysics — Explosive transients from massive stars: *hydrogen-poor superluminous supernovae (SLSNe-I)*, *bumpy SLSNe*, and *SNe with circumstellar interaction*. Tidal disruption events (TDEs) and stellar disruptions by massive black holes. Gamma-ray burst (GRB) afterglows and orphan afterglows. Electromagnetic counterparts of gravitational-wave events and kilonovae. Fast Blue Optical Transients (FBOTs) and other unusual short-duration transients. Multi-wavelength (UV – optical – NIR) follow-up of transients with ground- and space-based facilities (HST, JWST, Gemini, Magellan, MMT, SOAR). Development of robotic-telescope, photometric and spectroscopic data-reduction pipelines, image-subtraction techniques, and machine-learning real/bogus classifiers. Building time-domain follow-up programs for the LSST, ULTRASAT, and Roman era.

Academic Positions

Sept 2023 – present **Postdoctoral Fellow in Astrophysics**, Center for Astrophysics | Harvard & Smithsonian, Cambridge, MA, USA.

Advisor: Prof. Edo Berger (Time-domain & multi-messenger group).

2023 – present **Junior Investigator**, NSF AI Institute for Artificial Intelligence & Fundamental Interactions (IAIFI), Cambridge, MA, USA.

April 2023 – Aug 2023 **Institute Postdoctoral Fellow**, Indian Institute of Technology Bombay (IIT Bombay), Mumbai, India.

Education

2018 – 2023 **Ph.D. in Astrophysics**, Indian Institute of Technology Bombay (IIT Bombay), Mumbai, India.

Thesis: “Studying optical counterparts of gravitational waves and gamma-ray burst afterglows with the GROWTH-India Telescope.”

Advisor: Prof. Varun Bhalerao. **Course CGPA:** 9.3/10.

2015 – 2017 **M.Sc. (Astrophysics and Nuclear Physics)** – 69.6%, University of Delhi, Delhi, India.

2012 – 2015 **B.Sc. (Hons.) Physics** – 90.3%, University of Delhi, Delhi, India.

Honors, Awards and Fellowships

2026 **ASI Zubin Kembhavi Award** (team award), Astronomical Society of India, awarded to the GROWTH-India Telescope team — ASI Meeting, IIT Guwahati.

2026	Dhananjay Joshi Fundamental Research Award (team award), awarded to the GROWTH-India Telescope team, Mumbai.
2024	Justice Oak Award for Outstanding Thesis in Astronomy , Astronomical Society of India — ASI Meeting, IIA Bangalore.
2023	Young Astronomer Award (Honorable Mention), Astronomical Society of India — ASI Meeting 2023, IIT Indore.
2022	Best Poster Award , 40th Annual Meeting of the Astronomical Society of India.
2020	New Discovery Award (Special Mention), 38th Annual Meeting of the Astronomical Society of India.
2018 – 2020	LSSTC Data Science Fellowship , LSST Corporation — 2-year international fellowship.
2017	GATE Scholarship , Government of India — for pursuing Doctoral studies.
2016	CSIR-NET Fellowship , Council of Scientific & Industrial Research, India.
2012 – 2017	INSPIRE / Central Sector Scholarship for College and University Students, Ministry of Education, Government of India.

Successful Observing Proposals (as PI / Co-I)

<i>HST / JWST</i>	James Webb Space Telescope : 2 accepted programs (PI / Co-I) — JWST-GO-11372, JWST-GO-12468. HST Cycle-32 proposal accepted (Co-I, ID 17991).
<i>Gemini N 8.1 m</i>	9 accepted proposals (PI / Co-I) since 2023: GN-2024A-Q-108, GN-2024B-Q-111, GN-2024B-Q-130, GN-2025A-Q-108, GN-2025A-Q-114, GN-2025A-Q-132, GN-2025A-Q-145, GN-2025A-Q-318, GN-2025B-Q-321.
<i>Gemini S 8.1 m</i>	11 accepted proposals (PI / Co-I) since 2023: GS-2024A-DD-106, GS-2024B-Q-106, GS-2024B-Q-108, GS-2024B-Q-124, GS-2025A-Q-108, GS-2025A-Q-112, GS-2025A-Q-133, GS-2025A-Q-321, GS-2025B-Q-227, GS-2026A-Q-233, GS-2026A-Q-325.
<i>MMT 6.5 m</i>	≥ 20 accepted proposals (PI / Co-I), 2024A – 2026A (SAO programs).
<i>Magellan 6.5 m</i>	Baade / Clay — 15 accepted proposals (PI) across 2024A – 2026A.
<i>SOAR 4.1 m</i>	Goodman spectrograph — 3-year program accepted as Co-I (2025B-366722).
<i>FLWO 1.2 m</i>	Tillinghast / KeplerCam — 5 accepted proposals (PI), 2024A – 2025B.
<i>HCT 2.0 m</i>	Himalayan Chandra Telescope, IIA — 9 accepted proposals (PI / Co-I), Cycles 2019–2022.

Publications

Summary. Author / co-author of **48 refereed publications** in leading international journals (*Nature*, *Nature Astronomy*, *ApJ*, *ApJL*, *MNRAS*, *AJ*, *A&A*, *PASP*); **8 publications as first author**. In addition, >200 *GCN Circulars*, *Astronomer’s Telegrams* and *Transient Name Server* classification reports (not listed here), and contributions to >100 *Minor Planet Electronic Circulars*. The complete, continuously updated list is on [NASA ADS](#) and [ORCID](#).

Selected First-Author and High-Impact Publications

1. **Kumar, H.**, et al. “SN 2024afav: A Superluminous Supernova with Multiple Light-curve Bumps and Spectroscopic Signatures of Circumstellar Interaction.” *ApJL*, 998, L3 (2026).
2. **Kumar, H.**, et al. “A Detection of Helium in the Bright Superluminous Supernova SN 2024rmj.” *ApJ*, 992, 122 (2025).
3. **Kumar, H.**, et al. “A Near-infrared Search for Helium in the Superluminous Supernova SN 2024ahr.” *ApJ*, 987, 127 (2025).

4. **Kumar, H.**, et al. “AT 2023vto: An Exceptionally Luminous Helium Tidal Disruption Event from a Massive Star.” *ApJL*, 974, L36 (2024).
5. **Kumar, H.**, et al. “India’s First Robotic Eye for Time-domain Astrophysics: The GROWTH-India Telescope.” *AJ*, 164, 90 (2022).
6. Farah, J., et al. (incl. **Kumar, H.**). “A Lense–Thirring precessing magnetar engine drives a superluminous supernova.” *Nature*, 651, 321 (2026).
7. Andreoni, I., et al. (incl. **Kumar, H.**). “A very luminous jet from the disruption of a star by a massive black hole.” *Nature*, 612, 430 (2022).
8. Ho, A. Y. Q., et al. (incl. **Kumar, H.**). “Minutes-duration optical flares with supernova luminosities.” *Nature* (2023).
9. Ahumada, T., et al. (incl. **Kumar, H.**). “Discovery and confirmation of the shortest gamma-ray burst from a collapsar.” *Nature Astronomy*, 5, 917 (2021).
10. Anand, S., et al. (incl. **Kumar, H.**). “Optical follow-up of the neutron-star–black-hole mergers S200105ae and S200115j.” *Nature Astronomy*, 5, 46 (2021).

Complete List of Refereed Publications

Newest first. First-author papers are marked with [First author].

1. Farah, J., et al. (incl. **Kumar, H.**). “A Lense–Thirring precessing magnetar engine drives a superluminous supernova.” *Nature*, 651, 321 (2026).
2. Swain, V., et al. (incl. **Kumar, H.**). “GRB 230204B: GIT Discovery of a Fast-Fading Afterglow Associated with an Energetic GRB from a Massive-Star Progenitor.” *ApJ*, 1000, 58 (2026).
3. Blanchard, P., et al. (incl. **Kumar, H.**). “Hydrogen-poor Superluminous Supernovae in the Nebular Phase: Spectral Diversity Due to Ejecta Ionization as a Probe of the Power Source.” *ApJ*, 999, 59 (2026).
4. **[First author] Kumar, H.**, et al. “SN 2024afav: A Superluminous Supernova with Multiple Light-curve Bumps and Spectroscopic Signatures of Circumstellar Interaction.” *ApJL*, 998, L3 (2026).
5. Farah, J., et al. (incl. **Kumar, H.**). “When IIb Ceases To Be: Bridging the Gap between IIb and Short-Plateau Supernovae.” *ApJ*, 998, 321 (2026).
6. Hammerstein, E., et al. (incl. **Kumar, H.**). “The Jetted Tidal Disruption Event AT 2022cmc: Connections to the Optical TDE Population and Spectral Sub-classes through Late-time Follow-up.” *ApJ*, 996, 143 (2026).
7. Franz, N., et al. (incl. **Kumar, H.**). “Optimizing Kilonova Searches: A Case Study of the Type IIb SN 2025ulz in the Localization Volume of the Low-significance Gravitational-Wave Event S250818k.” *ApJL*, 994, L45 (2025).
8. **[First author] Kumar, H.**, et al. “A Detection of Helium in the Bright Superluminous Supernova SN 2024rmj.” *ApJ*, 992, 122 (2025).
9. Roychowdhury, T., et al. (incl. **Kumar, H.**). “Efficacy of galaxy catalogues for following up gravitational-wave events.” *JApA*, 46, 68 (2025).
10. Aamer, A., et al. (incl. **Kumar, H.**). “The Type I Superluminous Supernova Catalogue – II. Spectroscopic evolution in the photospheric phase, velocity measurements, and constraints on diversity.” *MNRAS*, 541, 2674 (2025).
11. Blanchard, P., et al. (incl. **Kumar, H.**). “JWST Observations of the Nearby, Precisely Localized FRB 20250316A: A Potential Near-IR Counterpart and Implications for FRB Progenitors.” *ApJL*, 989, L49 (2025).
12. **[First author] Kumar, H.**, et al. “A Near-infrared Search for Helium in the Superluminous Supernova SN 2024ahr.” *ApJ*, 987, 127 (2025).
13. Li, A., et al. (incl. **Kumar, H.**). “The Nature of Optical Afterglows without Gamma-Ray Bursts: Identification of AT 2023lcr and Multiwavelength Modeling.” *ApJ*, 985, 124 (2025).
14. Srinivasaragavan, G., et al. (incl. **Kumar, H.**). “Multiwavelength analysis of AT 2023sva: a luminous orphan afterglow with evidence for a structured jet.” *MNRAS*, 538, 351 (2025).

15. Perley, D., et al. (incl. **Kumar, H.**). “The luminous, slow-rising orphan afterglow AT 2019pim as a candidate moderately relativistic outflow.” *MNRAS*, 537, 2362 (2025).
16. Bolin, B., et al. (incl. **Kumar, H.**). “The Palomar twilight survey of ‘Ayló’chaxnim, Atiras, and comets.” *Icarus*, 425, 116333 (2025).
17. Ahumada, T., et al. (incl. **Kumar, H.**). “Searching for Gravitational-Wave Optical Counterparts with the Zwicky Transient Facility: Summary of O4a.” *PASP*, 136, 114201 (2024).
18. Gomez, S., et al. (incl. **Kumar, H.**). “The Type I Superluminous Supernova Catalogue – I. Light-curve properties, models, and catalogue description.” *MNRAS*, 535, 471 (2024).
19. [First author] **Kumar, H.**, et al. “AT 2023vto: An Exceptionally Luminous Helium Tidal Disruption Event from a Massive Star.” *ApJL*, 974, L36 (2024).
20. Basu, J., et al. (incl. **Kumar, H.**). “Multiwavelength Observations of Multiple Eruptions of the Recurrent Nova M31N 2008-12a.” *ApJ*, 966, 44 (2024).
21. Anand, S., et al. (incl. **Kumar, H.**). “Collapsars as Sites of r -process Nucleosynthesis: Systematic Photometric Near-infrared Follow-up of Type Ic-BL Supernovae.” *ApJ*, 962, 68 (2024).
22. Sicardy, B., et al. (incl. **Kumar, H.**). “Constraints on the evolution of the Triton atmosphere from occultations: 1989–2022.” *A&A*, 682, L24 (2024).
23. Srinivasaragavan, G., et al. (incl. **Kumar, H.**). “Characterizing the Ordinary Broad-Line Type Ic SN 2023pel from the Energetic GRB 230812B.” *ApJL*, 960, L18 (2024).
24. Ho, A. Y. Q., et al. (incl. **Kumar, H.**). “Minutes-duration optical flares with supernova luminosities.” *Nature* (2023).
25. Sharma, K., et al. (incl. **Kumar, H.**). “Astreaks: astrometry of NEOs with trailed background stars.” *MNRAS*, 524, 2651 (2023).
26. Teja, R., et al. (incl. **Kumar, H.**). “Far-ultraviolet to Near-infrared Observations of SN 2023ixf: A High-energy Explosion Engulfed in Complex Circumstellar Material.” *ApJL*, 954, L12 (2023).
27. Srinivasaragavan, G., et al. (incl. **Kumar, H.**). “A Sensitive Search for Supernova Emission Associated with the Extremely Energetic and Nearby GRB 221009A.” *ApJL*, 949, L39 (2023).
28. Andreoni, I., et al. (incl. **Kumar, H.**). “A very luminous jet from the disruption of a star by a massive black hole.” *Nature*, 612, 430 (2022).
29. Singhal, A., et al. (incl. **Kumar, H.**). “Gravitational waves and electromagnetic transients.” *JApA*, 43, 53 (2022).
30. [First author] **Kumar, H.**, et al. “GROWTH on S190426c – II: GROWTH-India Telescope search for an optical counterpart with a custom image-reduction and candidate-vetting pipeline.” *MNRAS*, 516, 4517 (2022).
31. Pearson, K., et al. (incl. **Kumar, H.**). “Utilizing a Global Network of Telescopes to Update the Ephemeris for the Highly Eccentric Planet HD 80606 b for Efficient Scheduling of JWST.” *AJ*, 164, 178 (2022).
32. Ho, A. Y. Q., et al. (incl. **Kumar, H.**). “Cosmological Fast Optical Transients with ZTF: A Search for Dirty Fireballs.” *ApJ*, 938, 85 (2022).
33. [First author] **Kumar, H.**, et al. “India’s First Robotic Eye for Time-domain Astrophysics: The GROWTH-India Telescope.” *AJ*, 164, 90 (2022).
34. [First author] **Kumar, H.**, et al. “The long-active afterglow of GRB 210204A: detection of the most delayed flares in a gamma-ray burst.” *MNRAS*, 513, 2777 (2022).
35. Ahumada, T., et al. (incl. **Kumar, H.**). “In Search of Short Gamma-Ray Burst Optical Counterparts with the Zwicky Transient Facility.” *ApJ*, 932, 40 (2022).
36. Gupta, R., et al. (incl. **Kumar, H.**). “Probing the emission mechanisms of GRB 190530A using time-resolved spectra and polarization studies: synchrotron origin?” *MNRAS*, 511, 1694 (2022).
37. Dutta, A., et al. (incl. **Kumar, H.**). “SN 2020sck: Deflagration in a Carbon–Oxygen White Dwarf.” *ApJ*, 925, 217 (2022).
38. Perley, D., et al. (incl. **Kumar, H.**). “Real-time discovery of AT 2020xnd: a fast, luminous UV

transient with minimal radioactive ejecta.” *MNRAS*, 508, 5138 (2021).

39. Andreoni, I., et al. (incl. **Kumar, H.**). “Fast-Transient Searches in Real Time with ZTFReST: Identification of Three Optically Discovered GRB Afterglows and New Constraints on the Kilonova Rate.” *ApJ*, 918, 63 (2021).
40. Ahumada, T., et al. (incl. **Kumar, H.**). “Discovery and confirmation of the shortest gamma-ray burst from a collapsar.” *Nature Astronomy*, 5, 917 (2021).
41. Misra, K., et al. (incl. **Kumar, H.**). “Low-frequency view of GRB 190114C reveals time-varying shock micro-physics.” *MNRAS*, 504, 5685 (2021).
42. Purdum, J., et al. (incl. **Kumar, H.**). “Time-series and Phase-curve Photometry of the Episodically Active Asteroid (6478) Gault in a Quiescent State using APO, GROWTH, P200, and ZTF.” *ApJL*, 911, L35 (2021).
43. Anand, S., et al. (incl. **Kumar, H.**). “Optical follow-up of the neutron-star–black-hole mergers S200105ae and S200115j.” *Nature Astronomy*, 5, 46 (2021).
44. Kasliwal, M. M., et al. (incl. **Kumar, H.**). “Kilonova Luminosity Function Constraints Based on ZTF Searches for 13 Neutron-Star Merger Triggers during O3.” *ApJ*, 905, 145 (2020).
45. Andreoni, I., et al. (incl. **Kumar, H.**). “GROWTH on S190814bv: Deep Synoptic Limits on the Optical/Near-IR Counterpart to a Neutron-Star–Black-Hole Merger.” *ApJ*, 890, 131 (2020).
46. Coughlin, M. W., et al. (incl. **Kumar, H.**). “GROWTH on S190425z: Searching Thousands of Square Degrees for an Optical/IR Counterpart to a Binary Neutron-Star Merger with ZTF and Palomar Gattini-IR.” *ApJL*, 885, L19 (2019).
47. Singh, A., et al. (incl. **Kumar, H.**). “SN 2018hna: A 1987A-like Supernova with a Signature of Shock Breakout.” *ApJL*, 882, L15 (2019).
48. Perley, D., et al. (incl. **Kumar, H.**). “The fast, luminous UV transient AT2018cow: extreme supernova, or disruption of a star by an intermediate-mass black hole?” *MNRAS*, 484, 1031 (2019).

Invited Talks, Seminars and Conference Presentations

Invited Colloquia and Seminars

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| 2025 – 26 | Colloquium speaker — Indian Institute of Astrophysics (IIA) Bangalore, Raman Research Institute (RRI) Bangalore, IUCAA Pune . “Hydrogen-poor super-luminous supernovae as probes of the deaths of the most massive stars.” |
| 2025 – 26 | Department Seminar — IIT Bombay, IIT Hyderabad, Indian Institute of Science (IISc) Bangalore. |
| 2025 | Invited speaker, IISER Mohali and IISER Pune — “Massive stars through the lens of hydrogen-poor super-luminous supernovae.” |
| 2024 | Invited talk (Best-Thesis Award lecture), 42nd Annual Meeting of the Astronomical Society of India, IIA Bangalore. |
| 2024 | Invited talk, Workshop 6 “Relevance of small-telescope networks in the era of ELTs”, ASI Annual Meeting. |
| 2023 | Invited talk, Indian Physics Association, IIT Roorkee. |
| 2021 | Invited (joint) talk with Prof. V. Bhalerao, National Science Day, IIT Bombay. |

Conference and Workshop Presentations

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| 2025 | Contributed talk, <i>Superluminous 2025: Superluminous Supernovae at the Dawn of New Surveys</i> , Cambridge, USA. |
| 2025 | Contributed talk, <i>VLT, SUPERNOVAE, BRUNO @65 & H₀ @?</i> , ESO Supernova Planetarium, Garching, Germany. |
| 2024 – 26 | Several research talks and journal-club presentations, Transient Tea, Center for Astrophysics Harvard. |

2023	Contributed talk, 41st Annual Meeting of the ASI, IIT Indore.
2022	E-talk, XXXIst General Assembly of the International Astronomical Union, Busan, South Korea.
2022	Invited talk, Zwicky Transient Facility (ZTF) Summer School.
2022	Poster, 40th Annual Meeting of the ASI.
2021	Contributed talk, 39th Annual Meeting of the ASI.
2020	Workshop, TOM Toolkit.
2018 – 20	LSSTC Data Science Fellowship Sessions 7–12.
2019	Tutor, GROWTH Astronomy School.
2018	GROWTH Annual Conference; First GROWTH Winter School (co-organiser), IIT Bombay; Summer School on Gravitational-Wave Astronomy, ICTS Bangalore; Posters, ASI Annual Meeting and Young Astronomers' Meet, PRL Ahmedabad.

Teaching Experience

- **Certified Carpentries Instructor** (Software / Data / Library Carpentries).
- **Teaching Assistant, Department of Physics, IIT Bombay** — courses include: PH556 Astrophysics II, PH426 Astrophysics I, PH434 Programming Lab, PH231 Electronics Lab I (Transistors), PH233 Electronics Lab II (Op-amp), PH230 Electronics Lab III (Digital), PH435/PH503 Microprocessors (2019, 2021), and PH443 M.Sc. Electronics Lab I; with Profs. Varun Bhalerao, Sai Vinjanampathy, Pradeep Sarin, Pramod Kumar, and Uma Shankar.
- **Tutor / Co-Instructor:** GROWTH Astronomy School (2019); First GROWTH Winter School — IIT Bombay (2018, co-organiser).

Research Mentoring of Students

Co-mentored students through GROWTH-India, CfA and IAIFI programs (Ph.D., M.S. and B.S. projects).

2018	Hrishikesh Iyer — Setup & automation of the GROWTH-India Telescope. <i>Now: Ph.D., UIUC.</i>
2018 – 19	Viraj Karambelkar — Photometry, setup & automation of GIT. <i>Now: Ph.D., Caltech; Hubble Fellow.</i>
2018 – 22	Gaurav Waratkar — GIT observations and EM–GW / GRB follow-up. <i>Ph.D. IIT Bombay; now at Caltech.</i>
2019	Vashistha Kochar, Debasmita Nandi, Abhinand V. — GIT data management, target visualisation, telescope/CCD characterisation.
2019 – 20	Maitreya Khandagale, Adeem Jasani (now M.S., U. Toronto), Vedant Shenoy — EM–GW & SNe follow-up, data processing, AstroSat–CZTI transients.
2020 – 21	Abhinav Jain (now SDE, Navi), Joy Sanghavi, Simran Joharle (Ph.D., MPIA / Heidelberg), Kritti Sharma (Ph.D., Caltech), Subham Samantaray, Kunal Deshmukh.
2021 – 22	Mohak Gulati, K. Aditya Krishna — GIT dashboard and database interface.
2022	Dhatri Dongre M., Anish Kalsi — Age estimation of Galactic globular clusters.
2021 – 23	Vishwajeet Swain — GIT observations, reduction and analysis across the science program.

2022 – 23	Aswin Suresh, Harsh Choudhary, Ravi Kumar, Tamojeet Roychowdhury, Ritwik Sharma, Dhananjay Raman — ML real/bogus classifiers, NEO astrometry, Fermi-GRB statistics, scheduling, auto-focus and dashboards for GIT.
2024 – 25	Karthik Yadavalli — FLWO 1.2 m photometric and image-subtraction pipeline.
2026 – present	Ayaan Bhatnagar — Correlations between magnetar spin-down parameters and bulk SLSN-I properties.
Jun 2026 – present	Ayushman Joshi — “Finding life in death”: search for precursor activity in bumpy superluminous supernovae.

Observing and Data-Reduction Experience

- **Gemini Observatory (N & S):** Photometric and spectroscopic data reduction; image subtraction. Reduced Gemini data contributed to the discovery of the shortest LGRB associated with a collapsar (Ahumada et al. 2021) and to first-author bumpy-SLSN and helium-search papers (Kumar H. et al. 2024, 2025a, 2025b, 2026).
- **MMT 6.5 m:** Observing; photometric and spectroscopic data reduction (Binospec, MMIRS). PI for SLSN searches, helium-detection programs in SLSNe, and gravitational-wave counterpart follow-up.
- **Magellan 6.5 m (Baade & Clay):** Observing; photometric and spectroscopic reduction (IMACS, LDSS3, FIRE). PI for SLSN searches and GW follow-up.
- **SOAR 4.1 m — Goodman:** Spectroscopic data reduction; long-program for SLSNe and TDE searches.
- **FLWO 1.2 m (KeplerCam):** Observing; photometric and spectroscopic reduction; PI for SLSN searches, GW counterpart follow-up and optical FRB counterpart searches.
- **GROWTH-India 0.7 m Telescope:** Automation and end-to-end data-reduction pipeline development; > 4 years of observing and photometric reduction.
- **2.0 m Himalayan Chandra Telescope (HCT):** Photometric and spectroscopic data reduction; image subtraction. Reduced data for multiple transients during my Ph.D.
- **Keck Observatory:** One-night remote observing experience from the Caltech remote-observing room.
- **3.6 m Devasthal Optical Telescope (DOT) and 1.3 m DFOT:** Data reduction; developed in-house photometric pipelines (DFOT pipeline used for GRB 210204A analysis).

Professional Service and Memberships

- **Panel reviewer**, U.S. National Science Foundation (NSF) — 2024.
- **Time Allocation Committee member**, NOIRLab community time (Gemini / SOAR / NEID) — 2024–2026.
- **Time Allocation Committee member**, 3.6 m Devasthal Optical Telescope (DOT), ARIES, India — 2021–2024.
- **Time Allocation Committee member**, Giant Metrewave Radio Telescope (GMRT), NCRA–TIFR — 2023.
- **Referee** for IOP Publishing journals (*ApJ*, *ApJL*, *PASP*) — 2023–present.
- **Organizer / Host**, Transient Tea weekly meeting, Center for Astrophysics | Harvard — 2024–2026.
- **Scientific Organizing Committee**, Young Astronomers’ Meet, ARIES Nainital — 2022.
- **GROWTH-India Telescope Development Team Leader**, IIT Bombay / IIA Bangalore — 2018–2022.
- **Coordinator**, Astro–Cosmo–Gravity Journal Club, IIT Bombay — 2018–2019.
- **Lifetime Member**, Astronomical Society of India — since 2022.

Technical Skills

- **Programming languages:** Python (proficient), C, C++, shell scripting, L^AT_EX, HTML, basic SQL.
- **Astronomy software:** Astropy, PyeIt, IRAF / PyRAF, SExtractor / SCAMP / SWarp / PSFEx,

ZOGY image subtraction, MCMC samplers (*emcee*, *dynesty*), HEALPix / healpy, ligo.skymap, SAOImage DS9, REDPIPE.

- **Pipelines built / co-built:** GROWTH-India Telescope photometry pipeline; FLWO 1.2 m photometry + image-subtraction pipeline (with K. Yadavalli); multi-instrument PyeIt-based spectroscopic reduction pipeline supporting Binospec / Goodman / LDSS3 (optical) and GNIRS / Flamingos-2 / MMIRS (NIR); in-house photometry pipelines for the 3.6 m DOT and 1.3 m DFOT.
- **Infrastructure:** Linux (Ubuntu / Debian), git / GitHub / GitLab, CI workflows, REST and broker APIs (TNS, Lasair, GCN, Slack, Twilio), TOM Toolkit deployment.
- **Languages:** English, Hindi, Sanskrit (reading).